Aspirin *Prodigy Anesthesia*

Aspirin is a non-steroidal anti-inflammatory agent that has gained much more attention in recent years for it's anti-platelet activity.

Salicylic acid is an organic acid that is rapidly absorbed in the stomach and upper small intestine and reaches peak plasma levels within 1-2 hours. It is hydrolyzed into acetic acid and salicylate by plasma esterases in the tissue and blood and it bound to albumin.

Aspirin irreversibly inhibits platelet cyclooxygenase so that the platelet is permanently inactivated. The duration of the antiplatelet activity, then, is 8-10 days, which is the lifespan of the platelet. The anti-inflammatory effects do not last as long as the tissues can produce more cyclooxygenase to replace the deactivated enzyme.

Aspirin use decreases the incidence of transient ischemic attacks, angina, myocardial infarction, and thrombosis after CABG surgery. Side effects include gastritis, gastric ulcers, duodenal ulcers, asthma, hepatotoxicity, renal toxicity, and gastrointestinal bleeding. Aspirin is contraindicated in patients with hemophilia. It has recently been proven valuable in the treatment of preeclampsia and eclampsia.

Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 638-640.

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